Practical 3

20% contribution to the final mark

7 questions, each has 5 points; in total $7 \times 5 = 35$ points

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Assignments

Q1 : Write an R code to compute the variogram for CD4 data and display the graph. Analyze the output.

Q2 : List some advantages and disadvantages of longitudinal data analysis.

Q3 : What is Toeplitz matrix structure for the covariance/correlation matrix? Provide the code of a simple example in R where you use this structure in any model you would like. I must be able to run your code and see the result.

Q4 : Verify (2.14).

Q5 : What are "wide" and "long" formats in longitudinal data analysis? Provide an example (real data or synthetic).

Q6 : Write an R code that can compute $\boldsymbol{D}$, $\boldsymbol{\Sigma}_i$ and $\boldsymbol{V}_i$ for Lab7-1.

Q7 : Consider the following model and do the whole analysis described in Labs 7 and 7-1.

$$\begin{aligned} \text{VISUAL}_{it} &= \beta_0 + \beta_1 \text{VISUAL0}_i + \beta_2 \text{TIME}_{it} + \beta_3 \text{TREAT}_i \\ &\quad + \beta_4 \text{TREAT}_i \times \text{TIME}_{it} \\ &\quad + b_{0i} + b_{2i} \text{TIME}_{it} + \epsilon_{it}. \end{aligned} \tag{2.24}$$